# Online Supplementary Material

## A Synergy of Quantum-Inspired Pathfinding Algorithms, Reinforcement Learning, and Graph Neural Networks

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This document holds the diagnostic, superseded and index tables that the thesis cites as Supplement S.1–S.5 and Tables S1–S11. Section, table, figure and equation numbers without an S refer to the thesis. Artefact paths are relative to `https://github.com/Amir-Shahriari/Thesis_Project`.

## S.1 Training-Budget Tiers and the Headline Scale

| **Metric (warm V1)** | 1× **(10 it)** | 4× **(40 it)** |
|---|---|---|
| Strict goal-reach rate | 22 / 25 (88 %) | **25 / 25 (100 %)** |
| Strict-win rate (warm < cold) | 21 / 25 (84 %) | **24 / 25 (96 %)** |
| Mean cost ratio vs Dijkstra | 17.3× | 14.8× |
| Median cost ratio vs Dijkstra | 10.5× | 4.3× |
| Cells at ≤ 5× Dijkstra | — | 13 / 25 |
| Cold strict goal-reach (reference) | 3 / 25 (12 %) | 2 / 25 (8 %) |

**Table S1:** 50 × 50 V1 warm results, 1× versus 4× tier, 5 seeds × 5 scenarios, batch 64. Training longer advances the perturbation schedule, so every across-tier comparison in this section is made on different realisations of each graph. Unmatched: all 22 comparable cells have a different Dijkstra reference, and the three cells unsolvable at 1× (`runs/eval_reachability_audit.json`) are connected at 4×. Reach supports no compute claim (three discordant against zero, McNemar $p = 0.25$); cost ratios are indicative only. Sources: `runs/sweep_50x50/aggregate_50x50.json`, `runs/sweep_50x50/sweep_v1_50x50_4x.json`.

| **Metric** | **V1 (warm, 100×100)** | **V1 (warm, 25×25, from §4.3.1)** |
|---|---|---|
| Strict goal-reach rate | **6 / 25 (24 %)** | 24 / 25 (96 %) |
| Reasonable-reach $\mathcal{R}_3$ | 0 / 25 (0 %) | 36 % |
| Mean composite cost (reached) | 2,248 ± 1,187 | 242 ± 197 |
| Mean cost ratio vs Dijkstra | 40.11 | 4.42 |
| Mean per-trajectory wall-clock (ms) | 99.4 ± 33.2 | 7.5 ± 10.5 |
| Strict-win rate (warm < cold) | 24 % | 84 % |
| Paired $t$-test $p$-value | untestable[§] | 0.0074 |

**Table S2:** V1 at 100 × 100 against the 25 × 25 baseline. Three cells are unsolvable at their evaluation states; over solvable cells warm reach is 6/22 (27.3 %). [§]Untestable, as at 50 × 50 under the reactive rule (§4.3.2): warm reaches 6 cells and cold 1, disjoint, so there are no pairs. The reach contrast is 6 discordant cells against 1, exact McNemar $p = 0.125$ (§4.3.3). Source: `runs/sweep_v1_on_100x100.json`.

| Seed | Scenario id | 1× reach | 4× reach | 4× cost | Cost ratio | Train (min) |
|---|---|---|---|---|---|---|
| 42 | seed191664964_s0 | No | Yes | 1,035 | 15.73 | 137.4 |
| 42 | seed191664964_s1 | No | No | — | — | 35.1 |
| 42 | seed191664964_s2 | No | Yes | 1,221 | 23.29 | 59.6 |
| 42 | seed191664964_s3 | No | No | — | — | 58.9 |
| 1337 | seed1173222464_s0 | No | Yes | 3,556 | 60.06 | 35.0 |
| 1337 | seed1173222464_s1 | No | Yes | 3,530 | 59.83 | 58.9 |
| 1337 | seed1173222464_s2 | No | No | — | — | 58.9 |
| 2024 | seed518677876_s1 | No | Yes | 4,638 | 82.88 | 35.1 |
| 2024 | seed518677876_s2 | No | No | — | — | 81.5 |
| 2024 | seed518677876_s4 | No | Yes | 776 | 15.22 | 181.4 |
| 7 | seed2029167941_s2 | No | Yes | 3,200 | 59.25 | 36.4 |
| 7 | seed2029167941_s3 | No | Yes | 579 | 9.08 | 124.2 |
| 7 | seed2029167941_s4 | No | No | — | — | 144.3 |
| 314159 | seed1830621171_s0 | No | Yes | 4,662 | 84.26 | 35.0 |
| 314159 | seed1830621171_s2 | No | Yes | 1,933 | 28.76 | 145.6 |
| **Aggregate** | | 0 / 15 (0 %) | **10 / 15 (66.7 %)** | — | mean 43.84× | mean 81.8 |
| *95 % binomial CI on 4× reach rate* | | [0.384, 0.882] | | | | |
| *McNemar's test (1× vs 4×)* | | $p = 0.0020$ | | | | |

**Table S3:** V1 long-train subset: 15 cells spanning all five seeds (two to four per seed), drawn from cells where the 1× budget failed; evaluated at iterations 10 and 40. The 4× instances are easier: 11 of 12 cells with a reference at both have a shorter optimum (sign test $p = 0.0063$; median $69.3 \rightarrow 60.0$), and one cell unsolvable at 1× is reached at 4×. The McNemar test is biased towards 4× by an unknown amount; the compute claim rests on the paired cold-start control. Sources: `runs/v1_long_train_subset_results.json`, `runs/v1_long_train_aggregate_15cell.json`.

## S.2 Encoder-Geometry Diagnostic

| **Measurement** | $25 \times 25$ ($d$=64) | $50 \times 50$ ($d$=128) |
|---|---|---|
| *(A) Embedding geometry* | | |
| IsoScore (1 = isotropic) | 0.044 | 0.034 |
| Effective rank | 20.1/64 | 25.6/128 |
| Mean pairwise cosine | 0.756 | 0.626 |
| after mean-centring | 0.021 | 0.019 |
| Locality $r_s$, encoder$^\dagger$ | 0.124 | 0.057 |
| Locality $r_s$, raw input features$^\dagger$ | 0.041 | 0.007 |
| Locality $r_s$, *ground-truth coordinates*$^\dagger$ | 0.091 | 0.075 |
| same probe, pure grid ($E_{\text{extra}}$=0) | 0.978 | 0.980 |
| *(B) Retrieval gate, by embedding treatment* | | |
| Saturation ($\text{sim}^{\max} > 0.95$), raw | 78.5 % | 56.5 % |
| centred | 12.3 % | 9.2 % |
| whitened | 0.05 % | 0.43 % |
| Retrieval quality, raw | 0.103 | 0.023 |
| centred | 0.102 | 0.025 |
| whitened | 0.090 | 0.022 |
| Gate AUC, raw | 0.584 | 0.636 |
| *(C) Metric probe: locality $r_s$ by readout*$^\dagger$ | | |
| Cosine (as deployed) | 0.130 | 0.054 |
| Learned ridge metric | 0.224 | 0.154 |
| Best learned metric (single best checkpoint) | 0.287 | 0.195 |

**Table S4:** Encoder-geometry diagnostics for §4.6.1, 10 checkpoints per scale (five cells × warm/cold). **(A)** Geometry, including locality (Spearman $r_s$, embedding against hop distance). **(B)** Retrieval gate under four treatments (raw, mean-centred, whitened, percentile-calibrated): saturation, gate AUC, and retrieval quality (1.0 retrieves the nearest stored state, 0.0 a random one). **(C)** Metric probe: a learned metric (ridge, MLP) on frozen embeddings predicting hop distance. Warm against cold at $25 \times 25$: IsoScore 0.059 against 0.029, effective rank 23.7 against 16.5, mean cosine 0.666 against 0.845, warm better in all five scenarios; five pairs cannot reach significance (minimum Wilcoxon $p = 0.0625$), and none replicates at $50 \times 50$ (2/5, 3/5, 3/5 scenarios; $p \geq 0.31$). No claim is made that warm-starting mitigates encoder collapse. At $25 \times 25$ whitening cuts saturation from 78.5 % to 0.05 % (Wilcoxon $p = 0.002$) with retrieval quality unchanged ($p = 0.28$; centring $p = 0.92$). $^\dagger$Uninterpretable because shortcut edges compress hop distance (§3.2), so no conclusion is drawn. Sources: `runs/geometry_study_aggregate.json`, `runs/geometry_audit.json`, `runs/retrieval_gate.json`, `runs/metric_probe.json`; positive control via `scripts/analysis/locality_control.py` with $E_{\text{extra}} = 0$, values in `runs/thesis_support/item_c_locality_control.json`.

## S.3 Cross-Distribution Transfer

| Scenario | Grid | $p_{\text{deact}}$ | V1 reach | V2 reach | V3 reach | V3 mean $\lambda^{\text{eff}}$ |
|---|---|---|---|---|---|---|
| in-dist | $100 \times 100$ | 0.30 | 2 / 10 | 3 / 10 | 2 / 10 | 0.321 |
| new-seed | $100 \times 100$ | 0.30 | 1 / 10 | 2 / 10 | 1 / 10 | 0.322 |
| low-pert | $100 \times 100$ | 0.15 | 2 / 10 | 2 / 10 | 3 / 10 | 0.312 |
| high-pert | $100 \times 100$ | 0.45 | 1 / 10 | 2 / 10 | 2 / 10 | 0.350 |
| small-grid | $50 \times 50$ | 0.30 | 1 / 10 | 1 / 10 | 2 / 10 | 0.337 |

**Table S5:** Cross-distribution transfer of V3 with the V1-trained agent and the V2 library, evaluated on five scenarios, ten queries per scenario (§4.5.2). V2 and V3, the expert library and the similarity gate are the superseded inference-time rules of Supplement S.4; inference-rule conditions are for cross-comparison only. Source: `runs/v3_transfer_benchmark.json`.

## S.4 Superseded Inference-Augmentation Strand

| Configuration | Agent | Library | $\lambda$ rule | Goal-reach |
|---|---|---|---|---|
| V1 (baseline, no library) | V1-trained | — | — | 11 / 25 (44 %) |
| V2-control (isolated mechanism effect) | V1-trained | yes | fixed 0.5 | 10 / 25 (40 %) |
| V2 published (agent + library, independent) | V2-trained | yes | fixed 0.5 | 8 / 25 (32 %) |

**Table S6:** V2 on the $100 \times 100$ sweep. Superseded and retained for cross-comparison only; no claim in this thesis rests on the tables of this section. Two training-free inference-time mechanisms were layered over the trained V1 agent at $100 \times 100$. V2 blends a persistent expert library's recommendation at a fixed $\lambda_{\text{retr}} = 0.5$; V3 replaces the fixed coefficient with a similarity gate, $\lambda(s) = \lambda_{\max} \cdot \text{sim}^{\max}(s)$, where $\text{sim}^{\max}(s)$ is the highest cosine similarity between the current node's encoder embedding and any state embedding stored in the library, clipped below at zero, and $\lambda^{\text{eff}}$ is the value of $\lambda(s)$ realised at a step, reported as a mean. V2-control uses the V1 checkpoint with the library at $\lambda_{\text{retr}} = 0.5$; V2 published uses a separately trained V2 agent. V1 and V2-control are at the matched 1,000-step budget (static regime of Table S10); V2 published comes from the 300-step headline sweep and is not budget-matched with them. Sources: `runs/unified_benchmark_100x100.json` (V1, static regime), `runs/sweep_v3_adaptive.json` (V2-control), `runs/sweep_v2_headline.json` (V2 published).

| Configuration | Goal-reach | Mean cost (reached) | Mean cost ratio | Mean $\lambda^{\text{eff}}$ |
|---|---|---|---|---|
| V1 (no library) | 11 / 25 (44 %) | $8{,}934 \pm 4{,}858$ | 168.67 | — |
| V2-control (fixed $\lambda = 0.5$) | 10 / 25 (40 %) | $7{,}388 \pm 5{,}118$ | 132.16 | 0.50 |
| V3 (adaptive $\lambda$, $\lambda_{\max} = 1.0$) | 12 / 25 (48 %) | $8{,}097 \pm 4{,}092$ | 127.78 | 0.365 |

**Table S7:** V3 against V2-control and V1; one V1 checkpoint and the V2 library, inference rule varied, all at the matched 1,000-step budget. V2 reaches 10/25 and V3 12/25, results the static regime of Table S10 reproduces cell for cell; the 6/25 of the 300-step V1 sweep, on different evaluation states, is not a matched baseline. At the matched budget neither mechanism separates from V1 ($b$=1, $c$=2, $p = 1.0$ for V2; $b$=3, $c$=2, $p = 1.0$ for V3), nor does V3 from V2 ($b$=3, $c$=1, $p = 0.63$). Route quality is poor in all three: 132.2× Dijkstra for V2 and 127.8× for V3 against V1's 168.7×. Sources: `runs/sweep_v3_adaptive.json` (V2-control, V3), `runs/unified_benchmark_100x100.json` (V1, static regime).

| $\lambda_{\max}$ | Goal-reach | Mean cost (reached) | Mean $\lambda^{\text{eff}}$ | Mean $\text{sim}^{\max}$ |
|---|---|---|---|---|
| 0.5 | 4 / 10 | 5,177 | 0.193 | 0.386 |
| 1.0 | **6 / 10** | 6,823 | 0.383 | 0.383 |
| 1.5 | 4 / 10 | 7,619 | 0.574 | 0.383 |
| 2.0 | 3 / 10 | 6,987 | 0.766 | 0.383 |

**Table S8:** V3 $\lambda_{\max}$ sensitivity on 10 cells (first two scenarios of each seed; not the first ten records of the sweep file). The $\lambda_{\max} = 1.0$ row is the headline run restricted to these cells, from `runs/sweep_v3_adaptive.json`; other rows from `runs/v3_lambda_ablation.json`.

| Mode | Reach | Mean cost (reached) | Mean cost ratio | Mean $\lambda^{\text{eff}}$ |
|---|---|---|---|---|
| V1 (no library) | 6/10 | 5,344 | 84.0 | — |
| V2 (fixed $\lambda_{\text{retr}} = 0.5$) | 7/10 | 6,649 | 116.6 | 0.500 |
| V3 (adaptive $\lambda$) | 4/10 | 5,595 | 94.6 | **0.974** |

**Table S9:** V3 on the 10-cell long-train subset, one long-trained V1 agent per cell, library built at the evaluation state (matched encoder). McNemar V1 versus V3: $b = 2, c = 4, p = 0.6875$. The gate does not detect novelty: here it saturates at $\lambda^{\text{eff}} = 0.974$ and V3 reaches 4/10 cells against V1's 6/10, so the unsaturated gate of the headline sweep (mean $\lambda^{\text{eff}} = 0.365$) reflects encoder-state drift between library construction and evaluation; the representation itself is measured in §4.6.1. The evaluation uses a different perturbation realisation from Table S3, where the same agents reach all ten cells. Source: `runs/v3_long_train_subset_results.json`.

| Regime (Δ) | Build | Goal-reach | Cost ratio | Latency (ms) | Survival |
|---|---|---|---|---|---|
| static | V1 | 11 / 25 (44 %) | 168.7 | 541 | — |
| | V2 | 10 / 25 (40 %) | 132.2 | 5,962 | — |
| | V3 | **12 / 25 (48 %)** | 127.8 | 5,011 | — |
| | A* | 24 / 25 (96 %) | 1.36 | 21.4 | — |
| | Cold | 2 / 25 (8 %) | 271.0 | 614 | — |
| $\Delta = 50$ | V1 | 11 / 25 | 165.3 | 612 | 1.00 |
| | V2 | 8 / 25 | 125.1 | 5,955 | 1.00 |
| | V3 | 11 / 25 | 130.1 | 6,227 | 1.00 |
| | A** | 24 / 25 | 1.36 | 27.3 | n/a |
| | Cold | 3 / 25 | 191.1 | 753 | 1.00 |
| $\Delta = 30$ | V1 | 11 / 25 | 208.8 | 629 | 1.00 |
| | V2 | 10 / 25 | 123.5 | 6,047 | 1.00 |
| | V3 | 10 / 25 | 138.9 | 6,696 | 1.00 |
| | A** | 24 / 25 | 1.36 | 20.9 | n/a |
| | Cold | 2 / 25 | 233.3 | 904 | 1.00 |
| $\Delta = 15$ | V1 | 13 / 25 | 161.7 | 763 | 1.00 |
| | V2 | 9 / 25 | 101.2 | 7,474 | 1.00 |
| | V3 | 11 / 25 | 190.2 | 7,682 | 1.00 |
| | A** | 24 / 25 | 1.36 | 15.8 | n/a |
| | Cold | 3 / 25 | 254.1 | 906 | 1.00 |

**Table S10:** Unified benchmark, $25 \times 5 \times 4 = 500$ evaluations (cells, conditions, regimes). Latency: mean per-trajectory wall-clock; survival: fraction of perturbation events after which the trajectory continued, 1 by construction (§3.7). *A* encounters zero perturbations in every regime (V1: 368–1,178) and its results are bit-identical across regimes, so its rows are only a reference for undisturbed path quality. Sources: `runs/unified_benchmark_100x100_aggregate.json`, `runs/unified_benchmark_100x100.json`.

# S.5 Source Artefacts

| Finding | Source artefact |
|---|---|
| 25×25 sweep | `runs/sweep_phase3_final.json` |
| 50×50 V1 sweep (1× tier, published June) | `runs/sweep_50x50/sweep_v1_50x50_1x.json` |
| Action-space A/B, reactive arm (50 × 50) | `runs/sweep_50x50_reactive_ctrl/` |
| Action-space A/B, masked arm (50 × 50) | `runs/sweep_50x50_masked_gpu/` |
| Action-space control, masked (100 × 100) | `runs/sweep_100x100_masked.json` |
| Masked multi-query control (50 × 50, 3 seeds) | `runs/fleet/reeval_seed*_50x50_mq_masked.json` |

| **Finding** | **Source artefact** |
|---|---|
| Per-run execution provenance ($50 \times 50$ A/B arms only) | `provenance.json` (in each arm's run directory) |
| 50×50 V1 sweep (4× tier) | `runs/sweep_50x50/sweep_v1_50x50_4x.json` |
| 50×50 aggregate (gates, CI; McNemar stored asymptotic, exact values recomputed from per-cell files) | `runs/sweep_50x50/aggregate_50x50.json` |
| 50×50 run record (config, S4-fix status) | `runs/sweep_50x50/REPORT.md` |
| Paired-test methods behind the gate scorecard | `runs/stats_reanalysis.json` |
| 100×100 V1 sweep | `runs/sweep_v1_on_100x100.json` |
| 100×100 V2 sweep + V2-control re-evaluation | `runs/sweep_v2_headline.json` (V2), `runs/sweep_v3_adaptive.json` (V2-control) |
| 100×100 V3 sweep | `runs/sweep_v3_adaptive.json` |
| $\lambda_{\max}$ ablation | `runs/v3_lambda_ablation.json` |
| Cross-distribution transfer benchmark | `runs/v3_transfer_benchmark.json` |
| V1 long-train single cell | `runs/v1_long_train_results.json` |
| V1 long-train 15-cell subset (per-cell records) | `runs/v1_long_train_subset_results.json` |
| V1 long-train 15-cell aggregate (statistics, CI, McNemar) | `runs/v1_long_train_aggregate_15cell.json` |
| V3 long-train single-cell (legacy) | `runs/v3_long_train_results.json` |
| V3 long-train 10-cell subset (per-cell records) | `runs/v3_long_train_subset_results.json` |
| V3 long-train 10-cell aggregate (statistics) | `runs/v3_long_train_subset_aggregate.json` |
| Within-trajectory perturbation (25×25, conditions A/B/C) | `runs/v3_realtime_25x25_aggregate.json` |
| Unified comparison | `runs/unified_benchmark_100x100.json`, `runs/unified_benchmark_100x100_aggregate.json` |
| Unified sanity check | `runs/unified_benchmark_sanity.json` |
| Pareto plot data | `runs/unified_pareto_100x100.csv` |
| V1 perturbed-copies check (P1 gate) | `runs/perturbation_phase4.json` |
| Cold-start 4× control (per-cell + aggregate) | `runs/cold_4x_control_results.json`, `runs/cold_4x_control_aggregate.json` |
| Reward-design ablation (30 runs) | `runs/reward_ablation/reward_ablation_results.json` |
| Reward-shaping control (five arms, $25 \times 25$) | `runs/shaping_control/aggregate.json`, `runs/shaping_control/lambda_*.json` |
| Eval-state reachability audit | `runs/eval_reachability_audit.json` |
| Consolidated provenance inventory | `runs/INVENTORY.md` |
| Demonstration-source ablation (25×25), run log | `runs/demo_source_ablation.log` |

| Finding | Source artefact |
|---|---|
| Demonstration-source ablation, partial join (9/50) | `runs/demo_source_ablation_partial.json` |
| Per-arm demonstration diversity (25×25) | `runs/traces_25x25/*/*/seeding_diversity.json` |
| Sample efficiency (episodes-to-threshold, 25×25) | `runs/learning_curves_25x25.json` |
| Multi-query training, per seed (25×25) | `runs/fleet/fleet_results_seed*_25x25.json` |
| Multi-query held-out re-evaluation (150 queries) | `runs/fleet/reeval_seed*_25x25.json` |
| Goal-relative head, per seed | `runs/fleet/reeval_seed*_25x25_gr.json` |
| Goal-relative across-seed aggregate | `runs/fleet/seed_grid_25x25_gr.json` |
| Hindsight experience replay (seed 42) | `runs/fleet/reeval_seed42_25x25_her.json` |
| Multi-query training, per seed (50 × 50) | `runs/fleet/fleet_results_seed*_50x50_mq.json` |
| Action-space A/B, reactive arm (25 × 25) | `runs/sweep_phase3_unmasked_ctrl.json` |
| Action-space A/B, masked arm (25 × 25) | `runs/sweep_phase3_masked.json` |
| Action-space A/B, both arms re-run on one recorded stack (25 × 25) | `runs/thesis_support/item_g_ab25_rerun/` |
| Termination-reason probe (reactive) | `runs/traces_reactive_probe/` |
| Masked replication, per-episode traces | `runs/traces_25x25_masked/` |
| Masked multi-query control (25 × 25, 5 seeds) | `runs/fleet/reeval_seed*_25x25_masked.json` |
| Encoder geometry, per checkpoint | `runs/geometry_audit.json` |
| Encoder geometry, aggregate (A/B/C studies) | `runs/geometry_study_aggregate.json` |
| Retrieval-gate treatments | `runs/retrieval_gate.json` |
| Learned-metric probe | `runs/metric_probe.json` |
| Pre-registered persistence ablation (50 × 50, masked) | `runs/persistence_ablation_50x50/aggregate.json`, `runs/persistence_ablation_50x50/cells_seed_*.json` |
| Persistence ablation pre-registration | `runs/persistence_ablation_50x50/PREREGISTRATION.md` |
| Demonstration-source ablation, traced full-pool and cold reference (25 × 25) | `runs/sweep_phase3_traced.json` |
| Multi-query held-out re-evaluation (50 × 50) | `runs/fleet/reeval_seed*_50x50_mq.json` |
| Exploration statistics on the 25 × 25 preset graph (regenerated) | `runs/thesis_support/item_a_preset_exploration_rules.json` |

| Finding | Source artefact |
|---|---|
| Within-trajectory Dijkstra-denominator inflation (regenerated) | `runs/thesis_support/item_b_realtime_dijkstra_inflation.json` |
| Locality positive control and hop IQR (regenerated) | `runs/thesis_support/item_c_locality_control.json` |
| Reward calibration and optimal returns (regenerated) | `runs/thesis_support/item_d_reward_calibration.json` |
| Exact graph diameters (regenerated) | `runs/thesis_support/item_e_exact_diameters.json` |
| Within-trajectory re-encode cost (regenerated) | `runs/thesis_support/item_f_reencode_timing.json` |

**Table S11:** Source artefacts, relative to the project repository. Rows marked (regenerated) were produced in October 2026: three from the recorded analysis scripts, and the Dijkstra-inflation, exact-diameter and re-encode-timing artefacts from new deterministic analyses, all kept in `runs/thesis_support/scripts/`. Artefacts from two working trees and successive code revisions are consolidated in `runs/`, indexed by `runs/INVENTORY.md`, which records origin and superseded or incomplete status. Only artefacts produced after the protocol of §3.8.1 carry a `provenance.json` (`torch` build, device, compute capability, revision); earlier stacks cannot be established retrospectively. Superseded and not cited: `runs/sweep_50x50_masked/` (stack unrecorded). Incomplete: the demonstration-source ablation's aggregation aborted on a reproducibility guard, and `runs/demo_source_ablation_partial.json` holds only 9 seed-42 cells from a separate re-run, whose quantum-only costs differ from the logged run on three of four cells; the figures of §4.5.1 are recovered from the run log by `scripts/analysis/ablation_from_log.py`. Other artefacts in the repository are not evidence: the superseded $50 \times 50$ masked run (Table B.4); and perturbation-level diagnostics, a multiplier sweep on demonstration checkpoints (five cells per scale) and three density-robust retraining attempts (one $50 \times 50$ cell), outside the evaluation grid and not reported in Chapter 4. The reward-shaping control (§4.4.2, Table B.8) is the one declared exception to the provenance rule.